EXPERIMENT 9

PERFORMANCE EVALUATION OF AM GENERATION AND DETECTION UNDER NOISY CHANNEL CONDITIONS USING MATLAB

OBJECTIVE:

i) To generate an Amplitude Modulated (AM) signal, introduce Additive White Gaussian Noise (AWGN), and analyze the effect of noise on the transmitted signal using MATLAB.

ii) To recover the original message signal from the noisy AM signal using envelope detection and compare the transmitted and recovered waveforms to evaluate the performance of the AM communication system.

PROGRAM:

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;

% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);

% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);

% Add AWGN Noise (10 dB SNR) without Toolbox
snr_db = 10;
```

```
signal_power = mean(am.^2);
noise_power = signal_power / (10^(snr_db/10));
noise = sqrt(noise_power)*randn(size(am));
noisy = am + noise;

% Envelope Detection
env = abs(hilbert(noisy));

% Display Results
figure;

subplot(5,1,1)
plot(t,m)
title('Message Signal')
grid on

subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on

subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on

subplot(5,1,4)
plot(t,noisy)
title('AM Signal with AWGN Noise')
```

```
grid on


subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
grid on
```

OUTPUT:

